

$$\frac{d\vec{y}}{dx} = \vec{f}(x, \vec{y})$$

$$x=0 \quad \vec{y}(0) = \vec{y}_{ini}$$

$$\frac{dT}{dt} = -h(T-U)$$

$$\frac{dU}{dt} = h(T-U) + q$$

$$t=0 \quad T(0) = T_{ini}$$

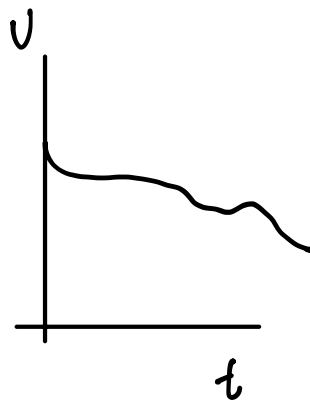
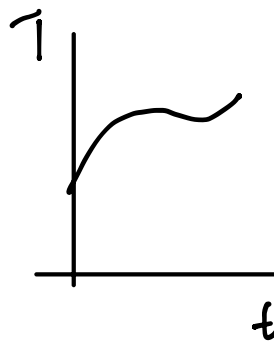
$$U(0) = U_{ini}$$

$$h, q \in \mathbb{R}.$$

$$U_{ini}, T_{ini} \in \mathbb{R}$$

$$T(t)$$

$$U(t)$$



$$\frac{dy}{dx} = y^2$$

$$\frac{dU}{dx} = x^2 + y$$

NOTACION VECTORIAL

$$\vec{y} = \begin{bmatrix} T \\ U \end{bmatrix}$$

$$\vec{f} = \begin{bmatrix} -h(T-U) \\ h(T-U) + q \end{bmatrix}$$

$$\frac{d\vec{y}}{dt} = \vec{f}(t, \vec{y})$$

$$t=0 \quad \vec{y} = \begin{bmatrix} T_{ini} \\ U_{ini} \end{bmatrix}$$

$$m \frac{d^2 x}{dt^2} = -gm$$

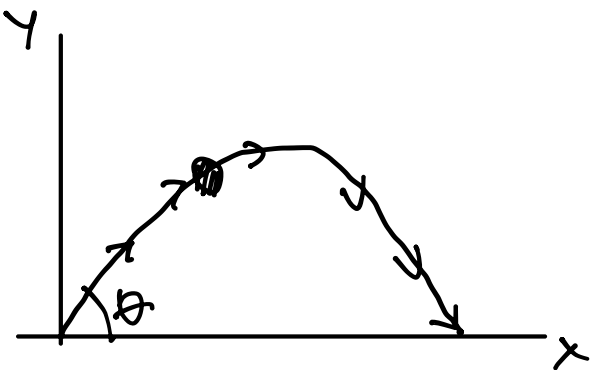
$$\frac{d^2 x}{dt^2} = -g$$

$$\begin{cases} \frac{dx}{dt} = v \\ \frac{dv}{dt} = -g \end{cases}$$

$$\vec{z} = \begin{bmatrix} x(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} z_0 \\ z_1 \end{bmatrix}$$

$$\vec{f} = \begin{bmatrix} v \\ -g \end{bmatrix} = \begin{bmatrix} f_0 \\ f_1 \end{bmatrix}$$

$$\frac{d\vec{z}}{dt} = \vec{f}(t, \vec{z})$$



$$m \frac{d^2 x}{dt^2} = 0$$

$$m \frac{d^2 y}{dt^2} = -gm$$

$$1) \frac{dx}{dt} = v$$

$$3) m \frac{dv}{dt} = 0$$

$$2) \frac{dy}{dt} = v$$

$$4) m \frac{dv}{dt} = -gm$$

$$\frac{dx}{dt} = v$$

$$\frac{dy}{dt} = v$$

$$\vec{z} = \begin{bmatrix} x \\ y \\ v \\ v \end{bmatrix} = \begin{bmatrix} z_0 \\ z_1 \\ z_2 \\ z_3 \end{bmatrix} \quad \vec{f} = \begin{bmatrix} v \\ v \\ 0 \\ -g \end{bmatrix} = \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ f_3 \end{bmatrix}$$

$$\frac{dv}{dt} = 0$$

$$\frac{dv}{dt} = -g$$

$$\frac{d\vec{z}}{dt} = \vec{F}(t, \vec{z})$$

$$y_{i+1} = y_i + h f(x_i, y_i)$$

$$\frac{dy}{dx} = f(x, y)$$

